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ITEM IDENTIFICATION USING DIGITAL IMAGE PROCESSING

Abstract

A device configured to detect a triggering event at a platform and to capture a depth image of items on the platform using a three-dimensional (3D) sensor. The device is further configured to determine an object pose for each item on the platform and to identify one or more cameras from among a plurality of cameras based on the object pose for each item on the platform. The device is further configured to capture one or more images of the items on the platform using the identified cameras and to identify items within the one or more images. The device is further configured to identify a user associated with the identified items on the platform, to identify an account that is associated with the user, and to associate the identified items with the account of the user.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 18/757,415, filed Jun. 27, 2024, which is a continuation of U.S. patent application Ser. No. 18/494,653, filed Oct. 25, 2023, now U.S. Pat. No. 12,094,165 issued Sep. 17, 2024, which is a continuation of U.S. patent application Ser. No. 17/362,261 filed Jun. 29, 2021, now U.S. Pat. No. 11,887,332 issued Jan. 30, 2024, which are all incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates generally to digital image processing, and more specifically to item identification using digital image processing.

BACKGROUND

[0003] Identifying and tracking objects within a space poses several technical challenges. Conventional systems may use various image processing techniques to identify objects. For example, these systems may identify different features of an item that can be used to later identify the item in an image. This process is computationally intensive when the image includes several items. For example, identifying an item in an image with multiple items involves identifying an individual item within the image and then comparing the features for an item against every item in a database of items. In addition to being computationally intensive, this process requires a significant amount of time which means that this process is not compatible with real-time applications. This problem becomes intractable when trying to simultaneously identify and track multiple items.

SUMMARY

[0004] The system disclosed in the present application provides a technical solution to the technical problems discussed above by using a combination of cameras and three-dimensional (3D) sensors to identify and track items that are placed on a platform. The disclosed system provides several practical applications and technical advantages which include a process for selecting a combination of cameras on an imaging device to capture images of items that are placed on a platform, identifying the items that are placed on the platform, and assigning the items to a user. Existing systems typically require a user to scan or manually identify items, which creates a bottleneck in the system's ability to quickly identify items. In contrast, the disclosed process is able to identify items from images of the items and assign the items to a user without requiring the user to scan or otherwise identify the items. This process provides a practical application by improving the system's ability to quickly identify multiple items. These practical applications not only improve the system's ability to identify items but also improve the underlying network and the devices within the network. For example, this disclosed process allows the system to service a larger number of users by reducing the amount of time that it takes to identify items and assign items to a user. In other words, this process improves hardware utilization without requiring additional

hardware resources which increases the number of hardware resources that are available for other processes and increases the throughput of the system. Additionally, these technical improvements allow for scaling of the item identification and tracking functionality described herein.

[0005] In one embodiment, the item tracking system comprises an item tracking device that is configured to detect a triggering event at a platform of an imaging device. The triggering event may correspond with when a user approaches or interacts with the imaging device by placing items on the platform. The item tracking device is configured to capture a depth image of items on the platform using a 3D sensor and to determine an object pose for each item on the platform based on the depth image. The pose corresponds with the location and the orientation of an item with respect to the platform. The item tracking device is further configured to identify one or more cameras from among a plurality of cameras on the imaging device based on the object pose for each item on the platform. This process allows the item tracking device to select the cameras with the best views of the items on the platform which reduces the number of images that are processed to identify the items. The item tracking device is further configured to capture images of the items on the platform using the identified cameras and to identify the items within the images based on features of the items. The item tracking device is further configured to identify a user associated with the identified items on the platform, to identify an account that is associated with the user, and to add the items to the account that is associated with the user.

[0006] Certain embodiments of the present disclosure may include some, all, or none of these advantages. These advantages and other features will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.

[0008] FIG. 1 is a schematic diagram of an embodiment of an item tracking system that is configured to employ digital image processing;

[0009] FIG. 2A is a perspective view of an embodiment of an imaging device;

[0010] FIG. 2B is a perspective view of another embodiment of an imaging device with an enclosure;

[0011] FIG. 2C is a perspective view of another embodiment of an imaging device with an open enclosure;

[0012] FIG. 3 is a flowchart of an embodiment of an item tracking process for the item tracking system;

[0013] FIG. 4 is an example of a comparison of depth images of a platform of the imaging device before and after placing items on the platform;

[0014] FIGS. 5A-5C are examples of images of items; and

[0015] FIG. 6 is an embodiment of an item tracking device configured for the item tracking system.

DETAILED DESCRIPTION

System Overview

[0016] FIG. 1 is a schematic diagram of an embodiment of an item tracking system **100** that is configured to employ digital image processing. The item tracking system **100** may employ digital image processing to identify items **204** that are placed on a platform **202** of an imaging device **102** and to assign the items **204** to a particular user. This process allows the user to obtain items **204** from a space without requiring the user to scan or otherwise manually identify the items **204** they would like to take. In one embodiment, the item tracking system **100** may be installed in a space

(e.g. a store) so that shoppers need not engage in the conventional checkout process. Although the example of a store is used in this disclosure, this disclosure contemplates that the item tracking system **100** may be installed and used in any type of physical space (e.g. a room, an office, an outdoor stand, a mall, a supermarket, a convenience store, a pop-up store, a warehouse, a storage center, an amusement park, an airport, an office building, etc.). As an example, the space may be a store that comprises a plurality of items **204** that are available for purchase. The item tracking system **100** may be installed in the store so that shoppers need not engage in the conventional checkout process to purchase items from the store. In this example, the store may be a convenience store or a grocery store. In other examples, the store may not be a physical building, but a physical space or environment where shoppers may shop. For example, the store may be a “grab-and-go” pantry at an airport, a kiosk in an office building, an outdoor market at a park, etc. As another example, the space may be a warehouse or supply room that comprises a plurality of items **204** that are available for a user to use or borrow. In this example, the item tracking system **100** may be installed to allow users to checkout parts or supplies by themselves. In other examples, the item tracking system **100** may be employed for any other suitable application.

[0017] In one embodiment, the item tracking system **100** comprises one or more imaging devices **102** and an item tracking device **104** that are in signal communication with each other over a network **106**. The network **106** may be any suitable type of wireless and/or wired network including, but not limited to, all or a portion of the Internet, an Intranet, a private network, a public network, a peer-to-peer network, the public switched telephone network, a cellular network, a local area network (LAN), a metropolitan area network (MAN), a personal area network (PAN), a wide area network (WAN), and a satellite network. The network **106** may be configured to support any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

Imaging Devices

[0018] The imaging device **102** is generally configured to capture images **122** and depth images **124** of items **204** that are placed on a platform **202** of the imaging device **102**. In one embodiment, the imaging device **102** comprises one or more cameras **108**, one or more three-dimensional (3D) sensors **110**, and one or more weight sensors **112**. Additional information about the hardware configuration of the imaging device **102** is described in FIGS. 2A-2C.

[0019] The cameras **108** and the 3D sensors **110** are each configured to capture images **122** and depth images **124** of at least a portion of the platform **202**. The cameras **108** are configured to capture images **122** (e.g. RGB images) of items **204**. Examples of cameras **108** include, but are not limited to, cameras, video cameras, web cameras, and printed circuit board (PCB) cameras. The 3D sensors **110** are configured to capture depth images **124** (e.g. depth maps or point clouds) of items. Examples of 3D sensors **110** include, but are not limited to, depth-sensing cameras, time-of-flight sensors, LiDARs, structured light cameras, or any other suitable type of depth sensing device. In some embodiments, a camera **108** and a 3D sensor **110** be integrated within a single device. In other embodiments, a camera **108** and a 3D sensor **110** be distinct devices.

[0020] The weight sensors **112** are configured to measure the weight of items **204** that are placed on the platform **202** of the imaging device **102**. For example, a weight sensor **112** may comprise a transducer that converts an input mechanical force (e.g. weight, tension, compression, pressure, or torque) into an output electrical signal (e.g. current or voltage). As the input force increases, the output electrical signal may increase proportionally. The item tracking device **104** is configured to analyze the output electrical signal to determine an overall weight for the items **204** on the weight sensor **112**. Examples of weight sensors **112** include, but are not limited to, a piezoelectric load cell or a pressure sensor. For example, a weight sensor **112** may comprise one or more load cells that are configured to communicate electrical signals that indicate a weight experienced by the load cells. For instance, the load cells may produce an electrical current that varies depending on the weight or force experienced by the load cells. The load cells are configured to communicate the produced electrical signals to item tracking device **104** for processing.

Item Tracking Device

[0021] Examples of the item tracking device **104** include, but are not limited to, a server, a computer, a laptop, a tablet, or any other suitable type of device. In FIG. **1**, the imaging device **102** and the item tracking device **104** are shown as two devices. In some embodiments, the imaging device **102** and the item tracking device **104** may be integrated within a single device. In one embodiment, the item tracking device **104** comprises an item tracking engine **114** and a memory **116**. Additional details about the hardware configuration of the item tracking device **104** are described in FIG. **6**. The memory **116** is configured to store item information **118**, user account information **120**, a machine learning model **126**, and/or any other suitable type of data.

[0022] In one embodiment, the item tracking engine **114** is generally configured to process images **122** and depth images **124** to identify items **204** that are placed on the platform **202** of the imaging device **102** and to associate the identified items **204** with a user. An example of the item tracking engine **114** in operation is described in more detail below in FIG. **3**.

[0023] The item information **118** generally comprises information that is associated with a plurality of items. Examples of item information **118** include, but are not limited to, prices, weights, barcodes, item identifiers, item numbers, features of items, or any other suitable information that is associated with an item **204**. Examples of features of an item include, but are not limited to, text, logos, branding, colors, barcodes, patterns, a shape, or any other suitable type of attributes of an item **204**. The user account information **120** comprises information for one or more accounts that are associated with a user. Examples of accounts include, but are not limited to, a customer account, an employee account, a school account, a business account, a financial account, a digital cart, or any other suitable type of account. The user account information **120** may be configured to associate user information with accounts that are associated with a user. Examples of user information include, but are not limited to, a name, a phone number, an email address, an identification number, an employee number, an alphanumeric code, reward membership information, or any other suitable type of information that is associated with the user. In some embodiments, the item information **118** and/or the user account information **120** may be stored in a device (e.g. a cloud server) that is external from the item tracking device **104**.

[0024] Examples of machine learning models **126** include, but are not limited to, a multi-layer perceptron, a recurrent neural network (RNN), an RNN long short-term memory (LSTM), a convolution neural network (CNN), a transformer, or any other suitable type of neural network model. In one embodiment, the machine learning model **126** is generally configured to receive an image **122** as an input and to output an item identifier based on the provided image **122**. The machine learning model **126** is trained using supervised learning training data that comprises different images **122** of items **204** with their corresponding labels (e.g. item identifiers). During the training process, the machine learning model **126** determines weights and bias values that allow the machine learning model **126** to map images **122** of items **204** to different item identifiers. Through this process, the machine learning model **126** is able to identify items **204** within an image **122**. The item tracking engine **114** may be configured to train the machine learning models **126** using any suitable technique as would be appreciated by one of ordinary skill in the art. In some embodiments, the machine learning model **126** may be stored and/or trained by a device that is external from the item tracking device **104**.

Hardware Configuration for the Imaging Device

[0025] FIG. **2A** is a perspective view of an embodiment of an imaging device **102**. In this example, the imaging device **102** comprises a platform **202**, a frame structure **206**, a plurality of cameras **108**, a plurality of 3D sensors **110**, and a weight sensor **112**. The imaging device **102** may be configured as shown in FIG. **2A** or in any other suitable configuration. In some embodiments, the imaging device **102** may further comprise additional components including, but not limited to, light, displays, and graphical user interfaces.

[0026] The platform **202** comprises a surface **208** that is configured to hold a plurality of items **204**.

In some embodiments, the platform **202** may be integrated with the weight sensor **112**. For example, the platform **202** may be positioned on the weight sensor **112** which allows the weight sensor **112** to measure the weight of items **204** that are placed on the platform **202**. As another example, the weight sensor **112** may be disposed within the platform **202** to measure the weight of items **204** that are placed on the platform **202**. In some embodiments, at least a portion of the surface **208** may be transparent. In this case, a camera **108** or scanner (e.g. a barcode scanner) may be disposed below the surface **208** of the platform **202** and configured to capture images **122** or scan the bottoms of items **204** placed on the platform **202**. For instance, a camera **108** or scanner may be configured to identify and read product labels and/or barcodes (e.g. stock-keeping units (SKUs)) of items **204** through the transparent surface **208** of the platform **202**. The platform **202** may be formed of aluminum, metal, wood, plastic, glass, or any other suitable material.

[0027] The frame structure **206** is generally configured to support and position cameras **108** and 3D sensors **110**. In FIG. 2A, the frame structure **206** is configured to position a first camera **108A** and a second camera **108C** on the sides of the imaging device **102** with a perspective view of the items **204** on the platform **202**. The frame structure **206** is further configured to position a third camera **108D** on the back side of the imaging device **102** with a perspective view of the items **204** on the platform **202**. In some embodiments, the frame structure **206** may further comprise a fourth camera **108** (not shown) on the front side of the imaging device **102** with a perspective view of items **204** on the platform **202**. The frame structure **206** may be configured to use any number and combination of the side cameras **108A** and **108C**, the back side camera **108D**, and the front side camera **108**. For example, one or more of the identified cameras **108** may be optional and omitted. A perspective image **122** or depth image **124** is configured to capture the side-facing surfaces of items **204** placed on the platform **202**. The frame structure **206** is further configured to position a third camera **108B** and a 3D sensor **110** with a top view or overhead view of the items **204** on the platform **202**. An overhead image **122** or depth image **124** is configured to capture upward-facing surfaces of items **204** placed on the platform **202**. In other examples, the frame structure **206** may be configured to support and position any other suitable number and combination of cameras **108** and 3D sensors **110**. The frame structure **206** may be formed of aluminum, metal, wood, plastic, or any other suitable material.

[0028] FIG. 2B is a perspective view of another embodiment of an imaging device **102** with an enclosure **210**. In this configuration, the enclosure **210** is configured to at least partially encapsulate the frame structure **206**, the cameras **108**, the 3D sensors **110**, and the platform **202** of the imaging device **102**. The frame structure **206**, the cameras **108**, the 3D sensors **110**, and the platform **202** may be configured similar to as described in FIG. 2A. In one embodiment, the frame structure **206** may further comprise rails or tracks **212** that are configured to allow the cameras **108** and the 3D sensors **110** to be repositionable within the enclosure **210**. For example, the cameras **108A**, **108C**, and **108D** may be repositionable along a vertical axis with respect to the platform **202** using the rails **212**. Similarly, camera **108B** and 3D sensor **110** may be repositionable along a horizontal axis with respect to the platform **212** using the rails **212**.

[0029] FIG. 2C is a perspective view of another embodiment of an imaging device **102** with an open enclosure **214**. In this configuration, the enclosure **214** is configured to at least partially cover the frame structure **206**, the cameras **108**, the 3D sensors **110**, and the platform **202** of the imaging device **102**. The frame structure **206**, the cameras **108**, the 3D sensors **110**, and the platform **202** may be configured similar to as described in FIG. 2A. In one embodiment, the frame structure **206** may be integrated within the enclosure **214**. For example, the enclosure **214** may comprise openings **216** that are configured to house the cameras **108** and the 3D sensors **110**. In FIG. 2C, the enclosure **214** has a rectangular cross section with rounded edges. In other embodiments, the enclosure **214** may be configured with any other suitable shape cross section.

An Item Tracking Process

[0030] FIG. 3 is a flowchart of an embodiment of an item tracking process **300** for the item

tracking system **100**. The item tracking system **100** may employ process **300** to identify items **204** that are placed on the platform **202** of an imaging device **102** and to assign the items **204** to a particular user. As an example, the item tracking system **100** may employ process **300** within a store to add items **204** to a user's digital cart for purchase. As another example, the item tracking system **100** may employ process **300** within a warehouse or supply room to check out items to a user. In other examples, the item tracking system **100** may employ process **300** in any other suitable type of application where items **204** are assigned or associated with a particular user. This process allows the user to obtain items **204** from a space without having the user scan or otherwise identify the items **204** they would like to take.

[0031] At step **302**, the item tracking device **104** performs auto-exclusion for the imaging device **102**. During an initial calibration period, the platform **202** may not have any items **204** placed on the platform **202**. During this period of time, the item tracking device **104** may use one or more cameras **108** and 3D sensors **110** to capture reference images **122** and reference depth images **124** of the platform without any items **204** placed on the platform **202**. The item tracking device **104** can then use the captured images **122** and depth images **124** as reference images to detect when an item is placed on the platform **202**. For example, the item tracking device **104** may use a 3D sensor **110** that is configured with a top view or overhead view of the platform **202** to capture a reference depth image **124** of the platform **202** when no items **204** are placed on the platform **202**. In this example, the captured depth image **124** may comprise a substantially constant depth value throughout the depth image **124** that corresponds with the surface **208** of the platform **202**. At a later time, the item tracking device **104** can detect that an item **204** has been placed on the surface **208** of the platform **202** based on differences in depth values between subsequent depth images **124** and the reference depth image **124**. As another example, the item tracking device **104** may use a camera **108** that is configured with a top view or a perspective view of the platform **202** to capture a reference image **122** of the platform when no items **204** are placed on the platform **202**. In this example, the captured image **122** comprises pixel values that correspond with a scene of the platform when no items **204** are present on the platform **202**. At a later time, the item tracking device **104** can detect that an item **204** has been placed on the platform **202** based on differences in the pixel values between subsequent images **122** and the reference image **122**.

[0032] At step **304**, the item tracking device **104** determines whether a triggering event has been detected. A triggering event corresponds with an event that indicates that a user is interacting with the imaging device **102**. For instance, a triggering event may occur when a user approaches the imaging device **102** or places an item **204** on the imaging device **102**. As an example, the item tracking device **104** may determine that a triggering event has occurred in response to detecting motion using a 3D sensor **110** or based on changes in depths images **124** captured by a 3D sensor **110**. For example, the item tracking device **104** can detect that an item **204** has been placed on the surface **208** of the platform **202** based on differences in depth values between depth images **124** captured by a 3D sensor **110** and the reference depth image **124**. Referring to FIG. **4** as an example, FIG. **4** shows an example of a comparison between depth images **124** from an overhead view of the platform **202** of the imaging device **102** before and after placing items **204** shown in FIG. **2A** on the platform **202**. Depth image **124A** corresponds with a reference depth image **124** that is captured when no items **204** are placed on the platform **202**. Depth image **124B** corresponds with a depth image **124** that is captured after items **204** are placed on the platform **202**. In this example, the colors or pixel values within the depth images **124** represent different depth values. In depth image **124A**, the depth values in the depth image **124A** are substantially constant which means that there are no items **204** on the platform **202**. In depth image **124B**, the different depth values correspond with the items **204** (i.e. items **204A**, **204B**, and **204C**) that are placed on the platform **202**. In this example, the item tracking device **104** detects a triggering event in response to detecting the presence of the items **204** on the platform **202** based on differences between depth image **124A** and depth image **124B**. The item tracking device **104** may also use an image **122** or depth image **124** to

count the number of items **204** that are on the platform **202**. In this example, the item tracking device **104** determines that there are three items **204** placed on the platform **202** based on the depth image **124B**. The item tracking device **104** may use the determined number of items **204** later to confirm whether all of the items **204** have been identified. This process is discussed in more detail below in step **312**.

[0033] As another example, the item tracking device **104** may determine that a triggering event has occurred in response to detecting motion using a camera **108** or based on changes in images **122** captured by a camera **108**. For example, the item tracking device **104** can detect that an item **204** has been placed on the platform **202** based on differences in the pixel values between subsequent images **122** and the reference image **122**. As another example, the item tracking device **104** may determine that a triggering event has occurred in response to a weight increase on the weight sensor **112** of the imaging device **102**. In this case, the increase in weight measured by the weight sensor **112** indicates that one or more items **204** have been placed on the platform **202**. In other examples, the item tracking device **104** may use any other suitable type of sensor or technique for detecting when a user approaches the imaging device **102** or places an item **204** on the imaging device **102**.

[0034] The item tracking device **104** remains at step **304** in response to determining that a triggering event has not been detected. In this case, the item tracking device **104** determines that a user has not interacted with the imaging device **102** yet. The item tracking device **104** will remain at step **304** to continue to check for triggering events until a user begins interacting with the imaging device **102**. The item tracking device **104** proceeds to step **306** in response to determining that a triggering event has been detected. In this case, the item tracking device **104** determines that a user has begun interacting with the imaging device **102**. The item tracking device **104** proceeds to step **306** to begin identifying items that are placed on the platform **202** of the imaging device **102**.

[0035] At step **306**, the item tracking device **104** identifies one or more cameras **108** for capturing images **122** of the items **204** on the platform **202** of the imaging device **102**. The item tracking device **104** may identify cameras **108** for capturing images **122** of the items **204** based at least in part upon the pose (e.g. location and orientation) of the items **204** on the platform **202**. The pose of an item **204** corresponds with the location the item **204** and how the item **204** is positioned with respect to the platform **202**. Referring to the example in FIG. 2A, a first item **204A** and a second item **204C** are positioned in a vertical orientation with respect to the platform **202**. In the vertical orientation, the identifiable features of an item **204** are primarily in the vertical orientation.

Cameras **108** with a perspective view, such as cameras **108A** and **108C**, may be better suited for capturing images **122** of the identifiable features of item **204** that are in a vertical orientation. For instance, the item tracking device **104** may select camera **108A** to capture images **122** of item **204A** since most of the identifiable features of item **204A**, such as branding, text, and barcodes, are located on the sides of the item **204A** and are most visible using a perspective view of the item **204**. Similarly, the item tracking device **104** may then select camera **108C** to capture images **122** of item **204C**. In this example, a third item **204B** is positioned in a horizontal orientation with respect to the platform **202**. In the horizontal orientation, the identifiable features of an item **204** are primarily in the horizontal orientation. Cameras **108** with a top view or overhead view, such as camera **108B**, may be better suited for capturing images **122** of the identifiable features of item **204** that are in a horizontal orientation. In this case, the item tracking device **104** may select camera **108B** to capture images **122** of item **204B** since most of the identifiable features of item **204B** are located on the top of the item **204B** and are most visible from using an overhead view of the item **204B**.

[0036] In one embodiment, the item tracking device **104** may determine the pose of items **204** on the platform **202** using depth images **124**. Referring to FIG. 4 as an example, the depth image **124B** corresponds with an overhead depth image **124** that is captured after the items **204** shown in FIG. 2A (i.e. items **204a**, **204B**, and **204C**) are placed on the platform **202**. In this example, the item tracking device **104** may use areas in the depth image **124B** that correspond with each item **204** to determine the pose of the items **204**. For example, the item tracking device **104** may determine the

area **402** within the depth image **124B** that corresponds with item **204A**. The item tracking device **104** compares the determined area **402** to a predetermined area threshold value **614**. The item tracking device **104** determines that an item **204** is in a vertical orientation when the determined area **402** for the item **204** is less than or equal to the predetermined area threshold value **614**. Otherwise, the item tracking device **104** determines that the item **204** is in a horizontal orientation when the determined area **402** for the item **204** is greater than the predetermined area threshold value **614**. In this example, the item tracking device **104** determines that items **204A** and **204C** are in a vertical orientation because their areas **402** and **406**, respectively, are less than or equal to the area threshold value **614**. The item tracking device **104** determines that item **204B** is in a horizontal orientation because its area **404** is greater than the area threshold value **614**. This determination means that the item tracking device **104** will select cameras **108** (e.g. cameras **108A** and **108C**) with a perspective view of the platform **202** to capture images **122** of items **204A** and **204C**. The item tracking device **104** will select a camera **108** (e.g. camera **108B**) with a top view or overhead view of the platform **202** to capture images **122** of item **204B**.

[0037] In one embodiment, the item tracking device **104** may identify a camera **108** for capturing images **122** of an item **204** based at least in part on the distance between the item **204** and the camera **108**. For example, the item tracking device **104** may generate homographies **608** between the cameras **108** and/or the 3D sensors **110** of the imaging device **102**. By generating a homography **608** the item tracking device **104** is able to use the location of an item **204** within an image **122** to determine the physical location of the item **204** with respect to the platform **202**, the cameras **108**, and the 3D sensors **110**. This allows the item tracking device **104** to use the physical location of the item **204** to determine distances between the item **204** and each of the cameras **108** and 3D sensors **110**. A homography **608** comprises coefficients that are configured to translate between pixel locations in an image **122** or depth image **124** and (x,y) coordinates in a global plane (i.e. physical locations on the platform **202**). The item tracking device **104** uses homographies **608** to correlate between a pixel location in a particular camera **108** or 3D sensor **110** with a physical location on the platform **202**. In other words, the item tracking device **104** uses homographies **608** to determine where an item **204** is physically located on the platform **202** based on their pixel location within an image **122** or depth image **124** from a camera **108** or a 3D sensor **110**, respectively. Since the item tracking device **104** uses multiple cameras **108** and 3D sensors **110** to monitor the platform **202**, each camera **108** and 3D sensor **110** is uniquely associated with a different homography **608** based on the camera's **108** or 3D sensor's **110** physical location on the imaging device **102**. This configuration allows the item tracking device **104** to determine where an item **204** is physically located on the platform **202** based on which camera **108** or 3D sensor **110** it appears in and its location within an image **122** or depth image **124** that is captured by that camera **108** or 3D sensor **110**. Additional information about generating a homography **608** and using a homography **608** is disclosed in U.S. Pat. No. 11,023,741 entitled, "DRAW WIRE ENCODER BASED HOMOGRAPHY" (attorney docket no. 090278.0233) which is hereby incorporated by reference herein as if reproduced in its entirety.

[0038] As an example, the item tracking device **104** may use an image **122** or a depth image **124** from a camera **108** or 3D sensor **110**, respectively, with a top view or overhead view of the platform **202** to determine the physical location of an item on the platform **202**. In this example, the item tracking device **104** may determine a pixel location for the item **204** within the image **122** or depth image **124**. The item tracking device **104** may then use a homography **608** to determine the physical location for the item **204** with respect to the platform **202** based on its pixel location. After determining the physical location of the item **204** on the platform **202**, the item tracking device **104** may then identify which camera **108** is physically located closest to the item **204** and select the identified camera **108**. Returning to the example in FIG. 2A, the item tracking device **104** may select camera **108A** to capture images **122** of item **204A** since camera **108A** is closer to item **204A** than camera **108C**. Similarly, the item tracking device **104** may select camera **108C** to capture

images **122** of item **204C** since camera **108C** is closer to item **204C** than camera **108A**. This process ensures that the camera **108** with the best view of an item **204** is selected to capture an image **122** of the item **204**.

[0039] At step **308**, the item tracking device **104** captures images **122** of the items **204** on the platform **202** using the identified cameras **108**. Here, the item tracking device **104** uses the identified cameras **108** to capture images of the items **204**. Referring to FIGS. 5A, 5B, and 5C as examples, the item tracking device **104** may capture a first image **122A** of the item **204A**, a second image **122B** of item **204B**, and a third image **122C** of item **204C** using cameras **108A**, **108B**, and **108C**, respectively. The item tracking device **104** may collect one or more images **122** of each item **204** for processing. By using a subset of the cameras **108** available on the imaging device **102** to capture images of the items **204**, the item tracking device **104** is able to reduce the number of images **122** that will be captured and processed to identify the items **204** on the platform **202**. This process reduces the search space for identifying items **204** and improves the efficiency and hardware utilization of the item tracking device **104** by allowing the item tracking device **104** to process fewer images **122** to identify the item **204** instead of processing images **122** from all of the cameras **108** on the imaging device **102**, which may include multiple images **122** of the same items **204**. In addition, the item tracking device **104** also selects cameras **108** that are positioned to capture features that are the most useful for identifying the items **204** based on the orientation and location of the items **204**, as discussed in step **306**. Examples of features include, but are not limited to, text, logos, branding, colors, barcodes, patterns, a shape, or any other suitable type of attributes of an item **204**.

[0040] Returning to FIG. 3 at step **310**, the item tracking device **104** identifies the items **204** on the platform **202** based on the captured images **122**. Here, the item tracking device **104** identifies an item **204** within each image **122** based on the features of the item **204** in the image **122**. As an example, the machine learning model **126** may be a CNN. In this example, the machine learning model **126** includes an input layer, an output layer, and one or more hidden layers. The hidden layers include at least one convolution layer. For example, the machine learning model **126** may include the following sequence of layers: input layer, convolution layer, pooling layer, convolution layer, pooling layer, one or more fully connected layers, output layer. Each convolution layer of machine learning model **126** uses a set of convolution kernels to extract features from the pixels that form an image **122**. In certain embodiments, the convolution layers of machine learning model **126** are implemented in the frequency domain, and the convolution process is accomplished using discrete Fourier transforms. This may be desirable to reduce the computational time associated with training and using machine learning model **126** for image classification purposes. For example, by converting to the frequency domain, the fast Fourier transform algorithm (FFT) may be implemented to perform the discrete Fourier transforms associated with the convolutions. Not only does the use of the FFT algorithm alone greatly reduce computational times when implemented on a single CPU (as compared with applying convolution kernels in the spatial domain), the FFT algorithm may be parallelized using one or more graphics processing units (GPUs), thereby further reducing computational times. Converting to the frequency domain may also be desirable to help ensure that the machine learning model **126** is translation and rotation invariant (e.g., the assignment made by machine learning model **126** of an image **122** to an item identifier, based on the presence of an item **204** in the image **122**, should not depend on the position and/or orientation of the item **204** within image **122**).

[0041] As another example, the machine learning model **126** may be a supervised learning algorithm. Accordingly, in certain embodiments, item tracking device **104** is configured to train the machine learning model **126** to assign input images **122** to any of a set of predetermined item identifiers. The item tracking device **104** may train the machine learning model **126** in any suitable manner. For example, in certain embodiments, the item tracking device **104** trains the machine learning model **126** by providing the machine learning model **126** with training data (e.g. images

122) that includes a set of labels (e.g. item identifiers) attached to the input images **122**. As another example, the machine learning model **126** may be an unsupervised learning algorithm. In such embodiments, the item tracking device **104** is configured to train machine learning model **126** by providing the machine learning model **126** with a collection of images **122** and instructing the machine learning model **126** to classify these images **122** with item identifiers identified by the item tracking device **104**, based on common features extracted from the images **122**. The item tracking device **104** may train the machine learning model **126** any time before inputting the captured images **122** into the machine learning model **126**.

[0042] After training the machine learning model **126**, the item tracking device **104** may input each of the captured images **122** into the machine learning model **126**. In response to inputting an image **122** in the machine learning model **126**, the item tracking device **104** receives an item identifier for an item **204** from the machine learning model **126**. The item identifier corresponds with an item **204** that was identified within the image **122**. Examples of item identifiers include, but are not limited to, an item name, a barcode, an item number, a serial number, or any other suitable type of identifier that uniquely identifies an item **204**.

[0043] In some embodiments, the item tracking device **104** may employ one or more image processing techniques without using the machine learning model **126** to identify an item **204** within an image **122**. For example, the item tracking device **104** may employ object detection and/or optical character recognition (OCR) to identify text, logos, branding, colors, barcodes, or any other features of an item **204** that can be used to identify the item **204**. In this case, the item tracking device **104** may process pixels within an image **122** to identify text, colors, barcodes, patterns, or any other characteristics of an item **204**. The item tracking device **104** may then compare the identified features of the item **204** to a set of features that correspond with different items **204**. For instance, the item tracking device **104** may extract text (e.g. a product name) from an image **122** and may compare the text to a set of text that is associated with different items **204**. As another example, the item tracking device **104** may determine a dominant color within an image **122** and may compare the dominant color to a set of colors that are associated with different items **204**. As another example, the item tracking device **104** may identify a barcode within an image **122** and may compare the barcode to a set of barcodes that are associated with different items **204**. As another example, the item tracking device **104** may identify logos or patterns within the image **122** and may compare the identified logos or patterns to a set of logos or patterns that are associated with different items **204**. In other examples, the item tracking device **104** may identify any other suitable type or combination of features and compare the identified features to features that are associated with different items **204**. After comparing the identified features from an image **122** to the set of features that are associated with different items **204**, the item tracking device **104** then determines whether a match is found. The item tracking device **104** may determine that a match is found when at least a meaningful portion of the identified features match features that correspond with an item **204**. In response to determining that a meaningful portion of features within an image **122** match the features of an item **204**, the item tracking device **104** may output an item identifier that corresponds with the matching item **204**. In other embodiments, the item tracking device **104** may employ one or more image processing techniques in conjunction with the machine learning model **126** to identify an item **204** within an image **122** using any combination of the techniques discussed above.

[0044] In some embodiments, the item tracking device **104** is configured to output a confidence score **610** that indicates a probability that an item **204** has been correctly identified. For example, the item tracking device **104** may obtain an confidence score **610** from the machine learning model **126** with the determined item identifier. In this example, the machine learning model **126** outputs a confidence score **610** that is proportional to the number of features that were used or matched when determining the item identifier. As another example, the item tracking device **104** may determine a confidence score **610** based on how well identified features match the features of the identified

item **204**. For instance, the item tracking device **104** may obtain a confidence score **610** of 50% when half of the text identified within an image **122** matches the text associated with identified item **204**. As another example, the item tracking device **104** may determine obtain a confidence score **610** of 100% when a barcode within an image **122** matches a barcode of the identified item **204**. As another example, the item tracking device **104** may obtain a confidence score **610** of 25% when the dominant color within an image **122** matches a dominant color of the identified item **204**. In other examples, the item tracking device **104** may obtain a confidence score **610** that is based on how well any other suitable type or combination of features matches the features of the identified item **204**. Other information that can impact a confidence score **610** include, but are not limited to, the orientation of the object, the number of items on the platform **208** (e.g., a fewer number of items on the platform **208** are easier to identify than a greater number of items on the platform **208**); the relative distance between items on the platform (e.g., spaced apart items on the platform **208** are easier to identify than crowded items on the platform **208**); and the like. The item tracking device **104** may compare the confidence score **610** for an identified item **204** to a confidence score threshold value **612** to determine whether the item **204** has been identified. The item tracking device **104** may determine that an item **204** has not been identified when the confidence score **610** for the item **204** is less than the confidence score threshold value **612**. The item tracking device **104** determines that the item **204** has been identified when the confidence score **610** for the item **204** is greater than or equal to the confidence score threshold value **612**. The confidence score threshold value **612** may be set to 90%, 80%, 75%, or any other suitable value.

[0045] At step **312**, the item tracking device **104** determines whether all of the items **204** on the platform **202** have been identified. For example, the item tracking device **104** may compare the number of identified items **204** from the captured images **122** to the number of items **204** on the platform **202** that was determined in step **304**. The item tracking device **104** determines that all of the items **204** on the platform **202** have been identified when the number of items **204** identified items **204** from the captured images **122** matches the determined number of items **204** on the platform **202**. Otherwise, the item tracking device **104** determines that at least one of the items **204** has not been identified when the number of items **204** identified items **204** from the captured images **122** does not match the determined number of items **204** on the platform **202**.

[0046] The item tracking device **104** proceeds to step **314** in response to determining that one or more of the items **204** on the platform **202** have not been identified. In this case, the item tracking device **104** may output a request for the user to reposition one or more items **204** on the platform **202** to assist the item tracking device **104** with identifying some of the items **204** on the platform. At step **314**, the item tracking device **104** outputs a prompt to rearrange one or more items **204** on the platform **202**. As an example, one or more items **204** may be obscuring the view of an item **204** for one of the cameras **108**. In this example, the item tracking device **104** may output a message on a graphical user interface that is located at the imaging device **102** with instructions for the user to rearrange the position of the items **204** on the platform **202**. In some embodiments, the item tracking device **104** may also identify the locations of the one or more items **204** on the platform **202** that were not identified. For example, the item tracking device **104** may activate a light source above or below the platform **202** that illuminates an item **204** that was not recognized. In one embodiment, after outputting the message to rearrange the items **204** on the platform **202**, the item tracking device **104** returns to step **306** to restart the process of identifying the items **204** on the platform **202**. This process prevents the item tracking device **104** from double counting items **204** after the items **204** have been rearranged on the platform **202**.

[0047] Returning to step **312**, the item tracking device **104** proceeds to step **316** in response to determining that all of the items **204** on the platform **202** have been identified. In some embodiments, the item tracking device **104** may validate the accuracy of detecting the identified items **204** based on the weight of the items **204** on the platform **202**. For example, the item tracking device **104** may determine a first weight that is associated with the weight of the identified items

204 based on item information **118** that is associated with the identified items **204**. For instance, the item tracking device **104** may use item identifiers for the identified items **204** to determine a weight that corresponds with each of the identified items **204**. The item tracking device **104** may sum the individual weights for the identified items **204** to determine the first weight. The item tracking device **104** may also receive a second weight for the items **204** on the platform **202** from the weight sensor **112**. The item tracking device **104** then determines a weight difference between the first weight and the second weight and compares the weight difference to a weight difference threshold value. The weight difference threshold value corresponds with a maximum weight difference between the first weight and the second weight. When the weight difference exceeds the weight difference threshold value, the item tracking device **104** may determine that there is a mismatch between the weight of the items **204** on the platform **202** of the imaging device **102** and the expected weight of the identified items **204**. In this case, the item tracking device **104** may output an error message and/or return to step **306** to restart the item tracking process. When the weight difference is less than or equal to the weight difference threshold value, the item tracking device **104** may determine that there is a match between the weight of the items **204** on the platform **202** of the imaging device **102** and the expected weight of the identified items **204**. In this case, the item tracking device **104** may proceed to step **316**.

[0048] At step **316**, the item tracking device **104** checks whether any prohibited or restricted item **204** are present on the platform **202**. A prohibited or restricted item **204** is an item **204** that the user is not authorized to obtain due to permission restrictions, age restrictions, or any other type of restrictions. The item tracking device **104** may compare item identifiers for the identified items **204** to a list of item identifiers for restricted or prohibited items **616**. In response to determining that an item **204** matches one of the items on the list of restricted or prohibited items **616**, the item tracking device **104** proceeds to step **318** to output an alert or notification that indicates that the user is prohibited from obtaining one of the items **204** that is on the platform **202**. For example, the item tracking device **104** may output an alert message that identifies the prohibited item **204** and asks the user to remove the prohibited item **204** from the platform **202** using a graphical user interface that is located at the imaging device **102**. As another example, the item tracking device **104** may output an alert message that identifies the prohibited item **204** to another user (e.g. an employee) that is associated with the space. In other examples, the item tracking device **104** may output any other suitable type of alert message in response to detecting a prohibited item **204** on the platform **202**.

[0049] At step **320**, the item tracking device **104** determines whether the prohibited item **204** has been removed from the platform **202**. For example, the item tracking device **104** may use the weight sensors **112** to determine whether the measured weight of the item **204** on the platform **202** has decreased by an amount that corresponds with the weight of the prohibited item **204**. As another example, the item tracking device **104** may use the cameras **108** and/or 3D sensors **110** to determine whether the prohibited item **204** is still present on the platform **202**. In response to determining that the prohibited item **204** is still present on the platform **202**, the item tracking device **104** may pause process **300** and remain at step **320** until the prohibited item **204** has been removed from the platform **202**. This process prevents the user from obtaining the prohibited item **204**. The item tracking device **104** may proceed to step **322** after the prohibited item **204** has been removed from the platform **202**.

[0050] Otherwise, the item tracking device **104** proceeds to step **322** in response to determining that no prohibited items **204** are present on the platform **202**. At step **322**, the item tracking device **104** associates the items **204** with the user. In one embodiment, the item tracking device **104** may identify the user that is associated with the items **204** on the platform **202**. For example, the user may identify themselves using a scanner or card reader that is located at the imaging device **102**. Examples of a scanner include, but are not limited to, a QR code scanner, a barcode scanner, a near-field communication (NFC) scanner, or any other suitable type of scanner that can receive an

electronic code embedded with information that uniquely identifies a person. In other examples, the user may identify themselves by providing user information on a graphical user interface that is located at the imaging device **102**. Examples of user information include, but are not limited to, a name, a phone number, an email address, an identification number, an employee number, an alphanumeric code, or any other suitable type of information that is associated with the user. [0051] The item tracking device **104** uses the information provided by the user to identify an account that is associated with the user and then to add the identified items **204** to the user's account. For example, the item tracking device **104** may use the information provided by the user to identify an account within the user account information **120** that is associated with the user. As an example, the item tracking device **104** may identify a digital cart that is associated with the user. In this example, the digital cart comprises information about items **204** that the user has placed on the platform **202** to purchase. The item tracking device **104** may add the items **204** to the user's digital cart by adding the item identifiers for the identified items **204** to the digital cart. The item tracking device **104** may also add other information to the digital cart that is related to the items **204**. For example, the item tracking device **104** may use the item identifiers to look up pricing information for the identified items **204** from the stored item information **118**. The item tracking device **104** may then add pricing information that corresponds with each of the identified items **204** to the user's digital cart.

[0052] After the item tracking device **104** adds the items **204** to the user's digital cart, the item tracking device **104** may trigger or initiate a transaction for the items **204**. In one embodiment, the item tracking device **104** may use previously stored information (e.g. payment card information) to complete the transaction for the items **204**. In this case, the user may be automatically charged for the items **204** in their digital cart when they leave the space. In other embodiments, the item tracking device **104** may collect information from the user using a scanner or card reader that is located at the imaging device **102** to complete the transaction for the items **204**. This process allows the items **204** to be automatically added to the user's account (e.g. digital cart) without having the user scan or otherwise identify the items **204** they would like to take. After adding the items **204** to the user's account, the item tracking device **104** may output a notification or summary to the user with information about the items **204** that were added to the user's account. For example, the item tracking device **104** may output a summary on a graphical user interface that is located at the imaging device **102**. As another example, the item tracking device **104** may output a summary by sending the summary to an email address or a user device that is associated with the user.

Hardware Configuration for the Item Tracking Device

[0053] FIG. **6** is an embodiment of an item tracking device **104** for the item tracking system **100**. In one embodiment, the item tracking device **104** may comprise a processor **602**, a memory **116**, and a network interface **604**. The item tracking device **104** may be configured as shown or in any other suitable configuration.

Processor

[0054] The processor **602** comprises one or more processors operably coupled to the memory **116**. The processor **602** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g. a multi-core processor), field-programmable gate array (FPGAs), application-specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **602** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The processor **602** is communicatively coupled to and in signal communication with the memory **116** and the network interface **604**. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **602** may be 8-bit, 16-bit, 32-bit, 64-bit, or of any other suitable architecture. The processor **602** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU

and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components.

[0055] The one or more processors are configured to implement various instructions. For example, the one or more processors are configured to execute item tracking instructions **606** to implement the item tracking engine **114**. In this way, processor **602** may be a special-purpose computer designed to implement the functions disclosed herein. In an embodiment, the item tracking engine **114** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The item tracking engine **114** is configured to operate as described in FIGS. **1** and **3**. For example, the item tracking engine **114** may be configured to perform the steps of process **300** as described in FIG. **3**.

Memory

[0056] The memory **116** is operable to store any of the information described above with respect to FIGS. **1** and **3** along with any other data, instructions, logic, rules, or code operable to implement the function(s) described herein when executed by the processor **602**. The memory **116** comprises one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **116** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM).

[0057] The memory **116** is operable to store item tracking instructions **606**, item information **118**, user account information **120**, machine learning models **126**, images **122**, depth images **124**, homographies **608**, confidence scores **610**, confidence score threshold values **612**, area threshold values **614**, a list of restricted or prohibited items **616**, and/or any other data or instructions. The item tracking instructions **606** may comprise any suitable set of instructions, logic, rules, or code operable to execute the item tracking engine **114**. The item information **118**, the user account information **120**, the machine learning models **126**, images **122**, depth images **124**, homographies **608**, confidence scores **610**, confidence score threshold values **612**, area threshold values **614**, and the list of restricted or prohibited items **616** are configured similar to the item information **118**, the user account information **120**, the machine learning models **126**, images **122**, depth images **124**, homographies **608**, confidence scores **610**, confidence score threshold values **612**, area threshold values **614**, and the list of restricted or prohibited items **616** described in FIGS. **1** and **3**, respectively.

Network Interface

[0058] The network interface **604** is configured to enable wired and/or wireless communications. The network interface **604** is configured to communicate data between the imaging device **102** and other devices, systems, or domains. For example, the network interface **604** may comprise an NFC interface, a Bluetooth interface, a Zigbee interface, a Z-wave interface, a radio-frequency identification (RFID) interface, a WIFI interface, a LAN interface, a WAN interface, a PAN interface, a modem, a switch, or a router. The processor **602** is configured to send and receive data using the network interface **604**. The network interface **604** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

[0059] While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated with another system or certain features may be omitted, or not implemented.

[0060] In addition, techniques, systems, subsystems, and methods described and illustrated in the

various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

[0061] To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 112(f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. An item tracking system, comprising: a plurality of cameras, wherein each camera is configured to capture images of at least a portion of a platform; and a three-dimensional (3D) sensor positioned above the platform, wherein the 3D sensor is configured to capture overhead depth images of items placed on the platform, wherein each overhead depth image represents surfaces of items placed on the platform; and a processor configured to: capture an overhead depth image of a first item using the 3D sensor; determine a first object pose for the first item based at least in part upon the overhead depth image, wherein determining the first object pose comprises determining a first orientation for the first item with respect to the platform; identify a first camera from among the plurality of cameras based at least in part upon the first object pose for the first item; capture a first image of the first item using the first camera; identify the first item within the first image; identify a user associated with the first item; identify an account associated with the user; and associate the first item with the account of the user.
2. The device of claim 1, further comprising a weight sensor configured to measure a weight for items on the platform, wherein the processor is further configured to detect a triggering event that comprises a weight increase on the weight sensor.
3. The device of claim 1, wherein the processor is further configured to detect a triggering event corresponding to the user placing at least one item on the platform by: capturing a second depth image of the platform; comparing the second depth image to a reference depth image, wherein the reference depth image is captured when no items are on the platform; and detecting the first item based on differences between the second depth image and the reference depth image.
4. The device of claim 1, wherein the processor is further configured to detect a triggering event corresponding to the user placing at least one item on the platform by: capturing an image of the platform; comparing the image of the platform to a reference image, wherein the reference image is captured when no items are on the platform; and detecting the first item based on differences between the image of the platform and the reference image.
5. The device of claim 1, further comprising a memory operable to store a homography associated with the 3D sensor, wherein the homography comprises coefficients that translate between pixel locations in the overhead depth image and physical locations on the platform; and wherein identifying the first camera from among the plurality of cameras comprises: applying the homography to the overhead depth image to determine a physical location of the first item on the platform; determining distances between the physical location of the first item and the plurality of cameras; and selecting the first camera from among the plurality of cameras that is closest to the physical location of the first item.
6. The device of claim 1, further comprising: a weight sensor configured to measure a weight for items on the platform; and a memory operable to store item information that comprises weights for a plurality of items; wherein the processor is further configured to: determine a first weight

associated with the first item based on the item information associated with the first item; receive a second weight for the first item from the weight sensor; determine a weight difference between the first weight and the second weight; determine the weight difference is less than or equal to a weight difference threshold value; and associate the first item with the account of the user in response to determining that the weight difference is less than or equal to a weight difference threshold value.

7. The device of claim 1, wherein identifying the first item within the first image comprises: inputting the first image into a machine learning model that is configured to output an item identifier based on features of an item that are present in an input image; and receiving a first item identifier for the first item in response to inputting the first image into the machine learning model.

8. The device of claim 1, wherein identifying the first camera from among the plurality of cameras based at least in part upon the first object pose comprises: determining the first item is in a vertical orientation with respect to the platform; and selecting the first camera from among the plurality of cameras in response to the determination, wherein the first camera is configured with a perspective view of the platform.

9. The device of claim 1, wherein identifying the first camera from among the plurality of cameras based at least in part upon the first object pose comprises: determining the first item is in a horizontal orientation with respect to the platform; and selecting the first camera from among the plurality of cameras in response to the determination, wherein the first camera is configured with an overhead view of the platform.

10. The device of claim 1, wherein determining the first object pose for the first item comprises: determining an area within the overhead depth image that corresponds with the first item; comparing the area to a predetermined area threshold value; determining that the area is less than or equal to the predetermined area threshold value; and selecting the first camera from among the plurality of cameras in response to the determination, wherein the first camera is configured with a perspective view of the platform.

11. The device of claim 1, wherein determining the first object pose for the first item comprises: determining an area within the overhead depth image that corresponds with the first item; comparing the area to a predetermined area threshold value; determining that the area is greater than the predetermined area threshold value; and selecting the first camera from among the plurality of cameras in response to the determination, wherein the first camera is configured with an overhead view of the platform.

12. The device of claim 1, wherein the processor is further configured to: compare item identifiers for the first item to a list of item identifiers for prohibited items; and determine that the item identifiers for the first item do not match any of the item identifiers for prohibited items before associating the first item with the account of the user.

13. The device of claim 1, wherein identifying the first item within the first image comprises: obtaining a confidence score that corresponds with a probability that the first item has been correctly identified; comparing the confidence score to a predetermined confidence score threshold value; and determining that the confidence score is greater than or equal to the confidence score threshold value.

14. The device of claim 1, further comprising a memory operable to store features for a plurality of items, wherein a feature corresponds with a physical attribute of an item; and wherein identifying the first item within the first image comprises: identifying features of the first item from the first image; comparing the identified features of the first item to the features for the plurality of items; identifying a second item from among the plurality of matches with features that at least partially match the identified features of the first item; and associating the first item with an item identifier that corresponds with the second item.

15. An item tracking method, comprising: detecting a triggering event at a platform, wherein the triggering event corresponds with a user placing items on the platform; capturing an overhead depth image of a first item using a three-dimensional (3D) sensor; determining a first object pose

for the first item based at least in part upon the overhead depth image, wherein determining the first object pose comprises determining a first orientation for the first item with respect to the platform; identifying a first camera from among a plurality of cameras based at least in part upon the first object pose for the first item; capturing a first image of the first item using the first camera; identifying the first item within the first image; identifying a user associated with the first item; identifying an account that is associated with the user; and associate the first item with the account of the user.

16. The method of claim 15, further comprising detecting a triggering event that comprises detecting a weight increase on a weight sensor, wherein the weight sensor is configured to measure a weight for items on the platform.

17. The method of claim 15, further comprising detecting a triggering event corresponding to the user placing at least one item on the platform by: capturing a second depth image of the platform; comparing the second depth image to a reference depth image, wherein the reference depth image is captured when no items are on the platform; and detecting first item based on differences between the second depth image and the reference depth image.

18. The method of claim 15, further comprising detecting a triggering event corresponding to the user placing at least one item on the platform by: capturing an image of the platform; comparing the image of the platform to a reference image, wherein the reference image is captured when no items are on the platform; and detecting the first item based on differences between the image of the platform and the reference image.

19. The method of claim 15, further comprising: determining a first weight associated with the first item based on item information associated with the first item; receiving a second weight for the first item from a weight sensor, wherein the weight sensor is configured to measure a weight for items on the platform; determining a weight difference between the first weight and the second weight; determining the weight difference is less than or equal to a weight difference threshold value; and associating the first item with the account of the user in response to determining that the weight difference is less than or equal to a weight difference threshold value.

20. The method of claim 15, wherein identifying the first item within the first image comprises: inputting the first image into a machine learning model that is configured to output an item identifier based on features of an item that are present in an input image; and receiving a first item identifier for the first item in response to inputting the first image into the machine learning model.

21. The method of claim 15, wherein identifying the first camera from among the plurality of cameras comprises: applying a homography to the overhead depth image to determine a physical location of the first item on the platform, wherein the homography comprises coefficients that translate between pixel locations in the overhead depth image and physical locations on the platform; determining distances between the physical location of the first item and the plurality of cameras; and selecting the first camera from among the plurality of cameras that is closest to the physical location of the first item.

22. The method of claim 15, wherein identifying the first camera from among the plurality of cameras based at least in part upon the first object pose comprises: determining the first item is in a vertical orientation with respect to the platform; and selecting the first camera from among the plurality of cameras in response to the determination, wherein the first camera is configured with a perspective view of the platform.

23. The method of claim 15, wherein identifying the first camera from among the plurality of cameras based at least in part upon the first object pose comprises: determining the first item is in a horizontal orientation with respect to the platform; and selecting the first camera from among the plurality of cameras in response to the determination, wherein the first camera is configured with an overhead view of the platform.

24. The method of claim 15, wherein determining the first object pose for the first item comprises: determining an area within the overhead depth image that corresponds with the first item;

comparing the area to a predetermined area threshold value; determining that the area is less than or equal to the predetermined area threshold value; and selecting the first camera from among the plurality of cameras in response to the determination, wherein the first camera is configured with a perspective view of the platform.

25. The method of claim 15, wherein determining the first object pose for the first item comprises: determining an area within the overhead depth image that corresponds with the first item; comparing the area to a predetermined area threshold value; determining that the area is greater than the predetermined area threshold value; and selecting the first camera from among the plurality of cameras in response to the determination, wherein the first camera is configured with an overhead view of the platform.

26. The method of claim 15, further comprising: comparing item identifiers for the first item to a list of item identifiers for prohibited items; and determining that the item identifiers for the first item do not match any of the item identifiers for prohibited items before associating the first item with the account of the user.

27. The method of claim 15, wherein identifying the first item within the first image comprises: obtaining a confidence score that corresponds with a probability that the first item has been correctly identified; comparing the confidence score to a predetermined confidence score threshold value; and determining that the confidence score is greater than or equal to the confidence score threshold value.

28. The method of claim 15, wherein identifying the first item within the first image comprises: identifying features of the first item from the first image; comparing the identified features of the first item to features for the plurality of items stored in a memory; identifying a second item from among the plurality of matches with features that at least partially match the identified features of the first item; and associating the first item with an item identifier that corresponds with the second item.

29. A non-transitory computer-readable medium storing instructions that when executed by a processor cause the processor to: detect a triggering event at a platform, wherein the triggering event corresponds with a user placing items on the platform; capture an overhead depth image of a first item using a three-dimensional (3D) sensor; determine a first object pose for the first item based at least in part upon the overhead depth image, wherein determining the first object pose comprises determining a first orientation for the first item with respect to the platform; identify a first camera from among a plurality of cameras based at least in part upon the first object pose for the first item; capture a first image of the first item using the first camera; identify the first item within the first image; identify a user associated with the first item; identify an account that is associated with the user; and associate the first item with the account of the user.

30. The non-transitory computer-readable medium of claim 29, wherein the instructions further cause the processor to detect a triggering event that comprises detecting a weight increase on a weight sensor, wherein the weight sensor is configured to measure a weight for items on the platform.

31. The non-transitory computer-readable medium of claim 29, wherein the instructions further cause the processor to detect a triggering event by: capturing a second depth image of the platform; comparing the second depth image to a reference depth image, wherein the reference depth image is captured when no items are on the platform; and detecting first item based on differences between the second depth image and the reference depth image.

32. The non-transitory computer-readable medium of claim 29, wherein the instructions further cause the processor to detect a triggering event by: capturing an image of the platform; comparing the image of the platform to a reference image, wherein the reference image is captured when no items are on the platform; and detecting the first item based on differences between the image of the platform and the reference image.

33. The non-transitory computer-readable medium of claim 29, wherein the instructions further

cause the processor to: determine a first weight associated with the first item based on item information associated with the first item; receive a second weight for the first item from a weight sensor, wherein the weight sensor is configured to measure a weight for items on the platform; determine a weight difference between the first weight and the second weight; determine the weight difference is less than or equal to a weight difference threshold value; and associate the first item with the account of the user in response to determining that the weight difference is less than or equal to a weight difference threshold value.

34. The non-transitory computer-readable medium of claim 29, wherein identifying the first item within the first image comprises: inputting the first image into a machine learning model that is configured to output an item identifier based on features of an item that are present in an input image; and receiving a first item identifier for the first item in response to inputting the first image into the machine learning model.

35. The non-transitory computer-readable medium of claim 29, wherein identifying the first camera from among the plurality of cameras comprises: applying a homography to the overhead depth image to determine a physical location of the first item on the platform, wherein the homography comprises coefficients that translate between pixel locations in the overhead depth image and physical locations on the platform; determining distances between the physical location of the first item and the plurality of cameras; and selecting the first camera from among the plurality of cameras that is closest to the physical location of the first item.

36. The non-transitory computer-readable medium of claim 29, wherein identifying the first camera from among the plurality of cameras based at least in part upon the first object pose comprises: determining the first item is in a vertical orientation with respect to the platform; and selecting the first camera from among the plurality of cameras in response to the determination, wherein the first camera is configured with a perspective view of the platform.

37. The non-transitory computer-readable medium of claim 29, wherein identifying the first camera from among the plurality of cameras based at least in part upon the first object pose comprises: determining the first item is in a horizontal orientation with respect to the platform; and selecting the first camera from among the plurality of cameras in response to the determination, wherein the first camera is configured with an overhead view of the platform.

38. The non-transitory computer-readable medium of claim 29, wherein determining the first object pose for the first item comprises: determining an area within the overhead depth image that corresponds with the first item; comparing the area to a predetermined area threshold value; determining that the area is less than or equal to the predetermined area threshold value; and selecting the first camera from among the plurality of cameras in response to the determination, wherein the first camera is configured with a perspective view of the platform.

39. The non-transitory computer-readable medium of claim 29, wherein determining the first object pose for the first item comprises: determining an area within the overhead depth image that corresponds with the first item; comparing the area to a predetermined area threshold value; determining that the area is greater than the predetermined area threshold value; and selecting the first camera from among the plurality of cameras in response to the determination, wherein the first camera is configured with an overhead view of the platform.

40. The non-transitory computer-readable medium of claim 29, wherein the instructions further cause the processor to: compare item identifiers for the first item to a list of item identifiers for prohibited items; and determine that the item identifiers for the first item do not match any of the item identifiers for prohibited items before associating the first item with the account of the user.

41. The non-transitory computer-readable medium of claim 29, wherein identifying the first item within the first image comprises: obtaining a confidence score that corresponds with a probability that the first item has been correctly identified; comparing the confidence score to a predetermined confidence score threshold value; and determining that the confidence score is greater than or equal to the confidence score threshold value.

42. The non-transitory computer-readable medium of claim 29, wherein identifying the first item within the first image comprises: identifying features of the first item from the first image; comparing the identified features of the first item to features for the plurality of items stored in a memory; identifying a second item from among the plurality of matches with features that at least partially match the identified features of the first item; and associating the first item with an item identifier that corresponds with the second item.
